The Newton Flow of The Hyperbolic Tangent Of One Plus t Squared

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Abstract

A closed form solution for the Newton flow of $\tanh(\ln(1+x^2))$ is derived and discussed in terms of it being the integrated infinitesimal generator comprising a semigroup of holomorphic mappings.

1 The Newton Flow ODE

The differential equation for the Newton flow $z(t)$ of $f(t)$ is given by

$$\dot{z}(t) = -\frac{f(z(t))}{\frac{d}{dt} f(z(t))} = -\frac{f(z(t))}{\dot{f}(z(t))} \tag{1}$$

If

$$f(t) = S(t) = \tanh\left(\ln\left(1+t^2\right)\right) \tag{2}$$

and we let

$$\begin{aligned} b(a) &= (1+a^2)^2+1 \\ &= a^4+2a^2+2 \end{aligned} \tag{3}$$

then define

$$\begin{aligned} g(t,a) &= e^t - 1 + \frac{2}{b(a)} \\ &= e^t - 1 + \frac{2}{(1+a^2)^2+1} \end{aligned} \tag{4}$$

and

$$\begin{aligned} h(t,a) &= \sqrt{e^{2t} - \frac{a^4\left(2+a^2\right)^2}{b(a)^2}} \\ &= \sqrt{e^{2t} - \frac{((1+a^2)^2-1)^2}{((1+a^2)^2+1)^2}} \end{aligned} \tag{5}$$

then there are 4 solutions of $z(t)$ for the Newton flow given by

$$\begin{aligned} z(t,a) &= \pm\sqrt{\pm\frac{g(t,a)+h(t,a)}{g(t,a)}} \\ &= \pm\sqrt{\pm\frac{e^t-1+\frac{2}{(1+a^2)^2+1}+\sqrt{e^{2t}-\frac{((1+a^2)^2-1)^2}{((1+a^2)^2+1)^2}}}{e^t-1+\frac{2}{(1+a^2)^2+1}}} \end{aligned} \tag{6}$$

which satisfy

$$\lim_{t\to\infty} z(t,a) = \{0, \pm i\sqrt{2}\} \forall a \in \mathbb{C} \tag{7}$$

where by design, definition, and necessity

$$S\Big(\lim_{t\to\infty} z(t,a)\Big) = 0 \forall a \in \mathbb{C} \tag{8}$$

1.1 Newton Flows Are Integrated Infinitesimal Generators Constituting Semigroups of Holomorphic Mappings

The Newton flow corresponding to the solution of the continuous Newton's method is the integrated infinitesimal generator constituting a semigroup of holomorphic mappings.

To understand this, we need to briefly discuss the continuous Newton's method and the concept of semigroups of holomorphic mappings.

Continuous Newton's method is a generalization of the classical Newton's method in which the iterations are replaced by a continuous flow. For a given holomorphic function $f(z)$, the continuous Newton's method is described by the following ordinary differential equation (ODE):

$$\frac{dz}{dt} = -\frac{f(z(t))}{f'(z(t))} \tag{9}$$

with an initial condition $z(0) = z_0$.

The solution of this ODE is a continuous trajectory $z(t)$ in the complex plane, which converges to the zero of the function $f(z)$ as t approaches infinity.

Definition 1

A `semigroup of holomorphic mappings` *is a family of mappings* $T(t): D \to D$*, where* D *is an open subset of the complex plane and* t *belongs to a certain interval* $[0, t_{\max})$*, with the following properties:*

1. $T(0)$ *is the identity map on* D*.*

2. *If* $s + t < t_{\max} \forall s, t \geq 0$ *then*

$$T(s) \circ T(t) = T(s+t) = T(\mathrm{sT}) \tag{10}$$

where $\circ$ *denotes the composition operator.*

Definition 2

An `infinitesimal generator` A *of a semigroup of holomorphic mappings* $T(t)$ *is a linear operator that satisfies the following condition:*

$$\lim_{t \to 0} \frac{T(t)\, z(t) - z(t)}{t} = A\, z(t) \forall z(t) \in D$$

Now, consider the Newton flow as a semigroup of holomorphic mappings $T(t)$ with an infinitesimal generator A. It can be shown that the Newton flow generated by the continuous Newton's method indeed satisfies the properties of a semigroup, and its infinitesimal generator A can be represented by the linear operator

$$-\frac{f(z)}{f'(z)} = \dot{z}(t) = -\frac{f(z(t))}{\frac{d}{dt} f(z(t))} = -\frac{f(z(t))}{\dot{f}(z(t))}$$

Here, the solution $z(t, a) = T(t)$ and the dependence on the initial condition a is still there just not written.

1.1.1 Conclusion

The Newton flow solution $z(t, a)$ is found by solving the ordinary Newton differential equation that describes how the flow evolves and its trajectory forms a semigroup under the operation of function composition.

The infinitesimal generator is essentially the velocity of the system, encoding its dynamics. When we talk about Newton flow, the infinitesimal generator is given by the right-hand side $-\frac{f(z(t))}{f'(z(t))}$ of the ODE:

$$\dot{z}(t) = -\frac{f(z(t))}{\frac{d}{dt} f(z(t))} = -\frac{f(z(t))}{\dot{f}(z(t))}$$

Here, $f(z) = S(z)$ is the function that we're trying to find roots for, and $f'(z)$ is its derivative. As time progresses, the points of the flow evolve according to the dynamics of the system. In the limit as t approaches infinity, $z(t, a)$ converges to a set of values $\{0, \pm i\sqrt{2}\}$, regardless of the initial codition $z(0) = a$ thus $S\left(\lim_{t\to\infty} z(t, a)\right) = 0$,

The connection between the infinitesimal generator and the semigroup properties of the flow is the fundamental principle that makes this method so powerful for root-finding.